1. recall that the Earth is surrounded by an atmosphere made up mainly of nitrogen, oxygen and argon, plus small amounts of water vapour, carbon dioxide, and other gases;
2. recall that the relative proportions of gases in the atmosphere are about 78% nitrogen, 21% oxygen and 1% argon;
3. recall that human activity adds small amounts of carbon monoxide, nitrogen oxides and sulfur dioxide to the atmosphere;
4. recall that human activity also adds extra carbon dioxide and small particles of solids (e.g. carbon) to the atmosphere;
5. recall that some of these substances, called pollutants, are directly harmful to humans and some are harmful to the environment and so cause harm to humans indirectly;
6. when using their own and given data relating to measured concentrations of atmospheric pollutants, or the composition of the atmosphere:
    - uses data rather than opinion in justifying an explanation;
    - can suggest reasons why a measurement may be inaccurate;
    - can suggest reasons why several measurements of the same quantity may give different results;
    - when asked to evaluate data, makes reference to its reliability (i.e. is it repeatable?);
    - can calculate the mean of a set of repeated measurements;
    - from a set of repeated measurements of a quantity, uses the mean as the best estimate of the true value;
    - can explain why repeating measurements leads to a better estimate of the quantity;
    - can make a sensible suggestion about the range within which the true value of a measured quantity probably lies;
    - **can justify the claim that there is/is not a ‘real difference’ between two measurements of the same quantity;**
    - can identify any outliers in a set of data, and give reasons for including or discarding them.